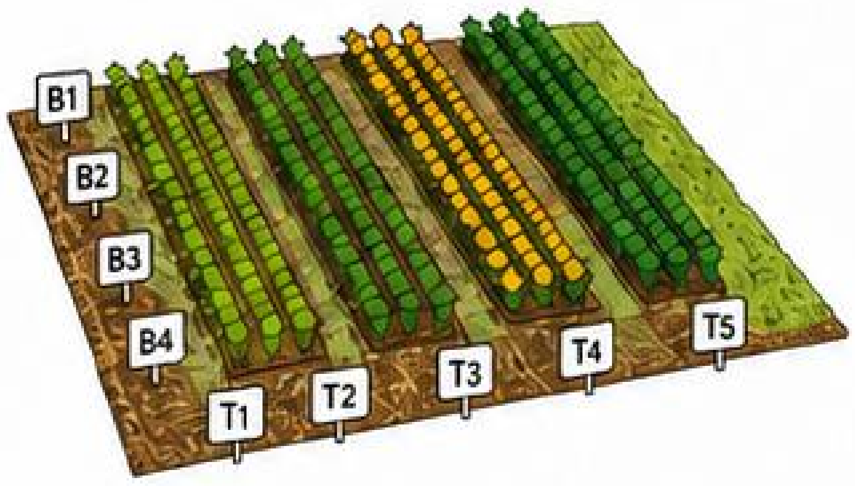
agriRank | Cheatsheet

Non-parametric inference for field trial design in agronomic experiments

1 What is agriRank?



- Declares an experiment before analysis
- Preserves randomization and strata
- Integrates tests, effects, contrasts, graphs, and reports
- Uses classic designs and modern modules

2 Workflow

Data → agri_design() → Validation → agri_rank() / agri_np_regression() → Effects and contrasts → Graphs → Report

3 Main modules

- RCBD/CRD
- LSD/ALPHA
- Factorial designs (A, B, C)
- Split-plot / Split-split / Strip-plot
- Repeated measures
- Missing data
- Conversion and comparisons
- Non-parametric regression
- Regression with integer support (0 1 2 3 4 5)
- Multivariate (PC1, PC2)
- Multienvironment
- Batch and sensitivity
- Plotly, tables and reports

4 Statistical principles

- Analyze as randomized
- Avoid choice by isolated p-value
- Effects and CIs with joint tests
- Comparative methods with discrete, ordinal, and unbalanced data (1 2 3 4 5)
- Reproducibility from import to report

5 Roadmap of this series

2. Designs
3. RCBD and CRD
4. Factorial designs
5. Repeated measures + missing
6. Effects and conversion
7. Multivariate + multienvironment
8. Graphs and reports
9. Regression
10. Regression with integer support (0 1 2 3 4 5)

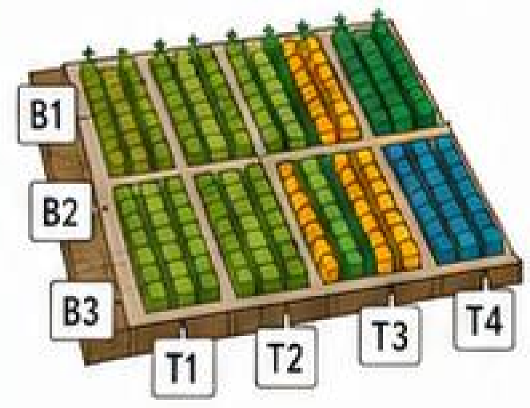
Declare an experiment → preserves inference → communicates with clarity. ***That's agriRank.***

agriRank | Experimental designs and declaration of the experiment

Analysis starts with experimental design

1 Why declare the design?

- Define an experimental unit
- Preserve blocks and strata
- Prevent incompatible analyses
- Guide the method and interpretation



2 agri_design()

DIC:	`agri_design(yield ~ treatment, data = dat, design = "crd")`
DBC:	`agri_design(yield ~ treatment, data = dat, design = "rcbd", block = block)`
Repeated:	`agri_design(height ~ treatment * time, data = dat, design = "repeated", subject = subject, within = time)`
Split-plot:	`agri_design(yield ~ irrigation * cultivar, data = dat, design = "split_plot", block = block, whole_plot = irrigation, subplot = cultivar)`

3 Covered experimental designs

- **CRD** — Homogeneous and independent units.
- **RCBD** (B1, B2, B3) — Blocks control known variability.
- **Factorial** (A, B, C) — Two or more factors crossed with each other.
- **Split-plot** — Factors in main plots and subplots.
- **Split-split** — Three levels of hierarchical plots.
- **Strip-plot** — Strip plots in the direction of gradients.
- **Repeated measures** (Time: t1 t2 t3 ... tn; Subject 1, 2, 3, ...) — Observations repeated on the same subject.
- **Multi-environment** — Trials in locations/years considered together.

4 Automatic validation

The `validate_agri_design()` function checks

Missing variables → Empty cells → Replication → Subject × time duplicated → Missing block → Numeric factor used as block → Integer support → Estimability ($X\beta = 0$ only trivial solution)

5 Key outputs

design_summary()
- Declared experimental structure
- Factors, levels and papers
- Hierarchy of plots and units

agri_methods()
- Compatible methods
- Degrees of freedom
- Available estimators

Warnings / alerts
- Incompatibilities
- Detected problems
- Correction recommendations

Randomization description
- Strata / blocks
- Allocation order
- Seed and date

6 Good practices

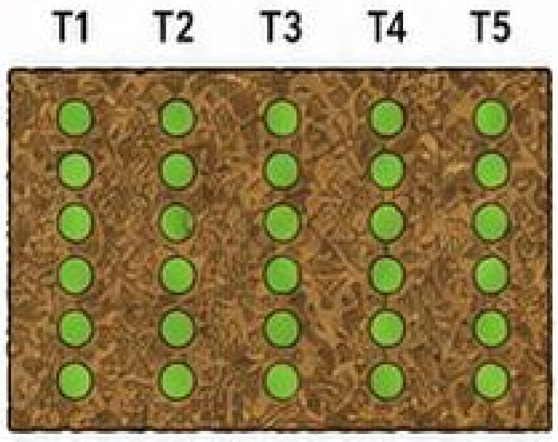
1. **Do not ignore blocks** — Always declare blocks/strata when they exist.
2. **Specify subject and within** (Subject; Time: t1 t2 t3 ... tn) — Essential for repeated measures and time series.
3. **Keep the scientific model** — $y \sim A * B + C$ (interaction, additive, covariate). Use interactions only when they have a biological meaning.
4. **Distinguish qualitative from quantitative treatment** — Qualitative (levels), Quantitative (gradient). Avoid misinterpretations and inappropriate methods.
5. **Review randomization before p-value** — Confirm the design and randomization before analysis.

Declare the experiment → preserve the inference → communicate with clarity. *That is agriRank.*

agriRank | CRD and RCBD

One-way nonparametric inference in completely randomized and randomized block designs

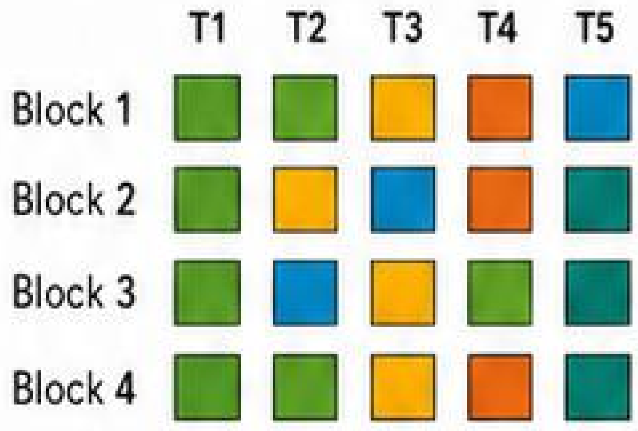
1 CRD / Completely Randomized Design



- independent treatments
- no blocking
- comparison of groups

`agri_design()` → `agri_rank()` `(method = "kruskal")` or `agri_rank()` `(method = "auto")`

RCBD / Randomized Complete Block Design



- complete blocks
- an observation per block × treatment in the classical model
- preserving blocks is mandatory

`agri_design()` → `agri_rank()` `(method = "friedman")` or `agri_rank()` `(method = "auto")`

2 Omnibus tests

Aspect	CRD (Completely Randomized Design)	RCBD (Randomized Complete Block Design)
Classical test	Kruskal–Wallis	Friedman
Recommended method	permutations and effects	Friedman or rank-based alternatives
Post-hoc	Conover, Dunn, etc.	Conover for Friedman when admissible
Common error	ignoring the block	ignoring the block

Essential formulas

Kruskal–Wallis (H)

$$H = \frac{12}{N(N+1)} \sum_{j=1}^{k} \frac{R_j^2}{n_j} - 3(N+1)$$

$$H \sim \chi^2_{k-1} \text{ (approx.)}$$

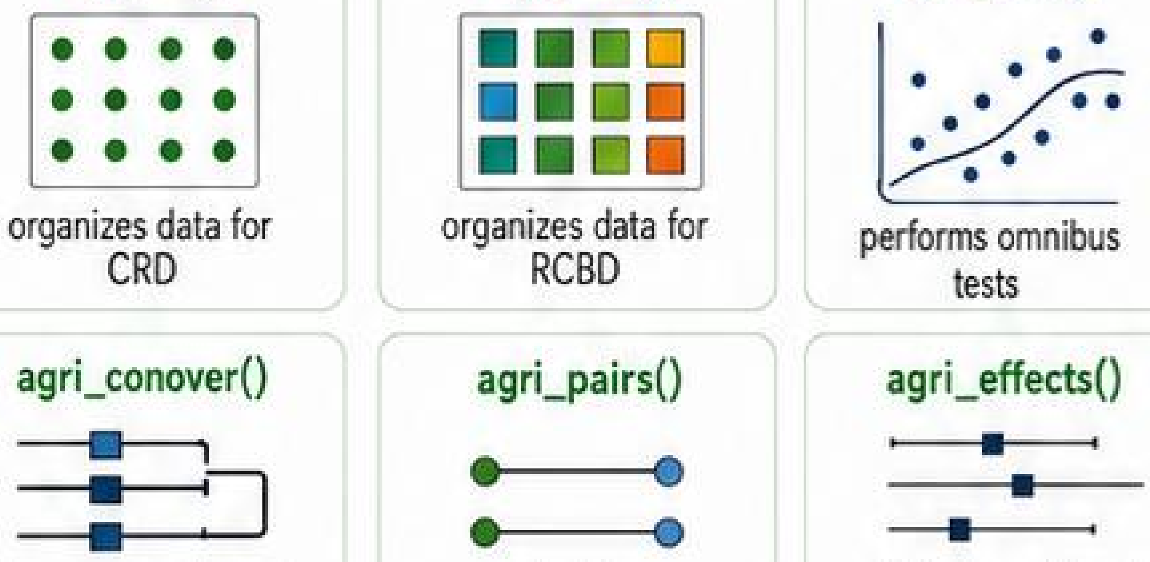
Friedman (χ^2_F)

$$\chi^2_F = \frac{12}{bk(k+1)} \sum_{j=1}^{k} R_j^2 - 3b(k+1)$$

$$\chi^2_F \sim \chi^2_{k-1} \text{ (approx.)}$$

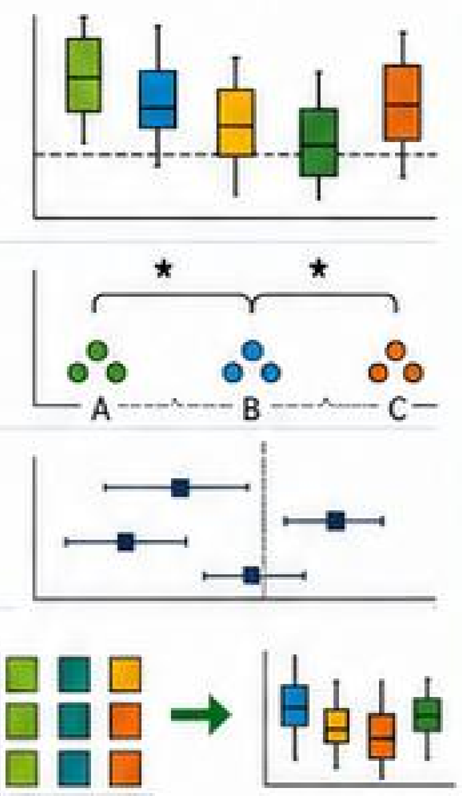
Where: k = number of treatments; $N = \sum_j n_j$ (CRD); b = number of blocks (RCBD); R_j = sum of treatment *ranks* j; n_j = size of group j.

3 Useful functions



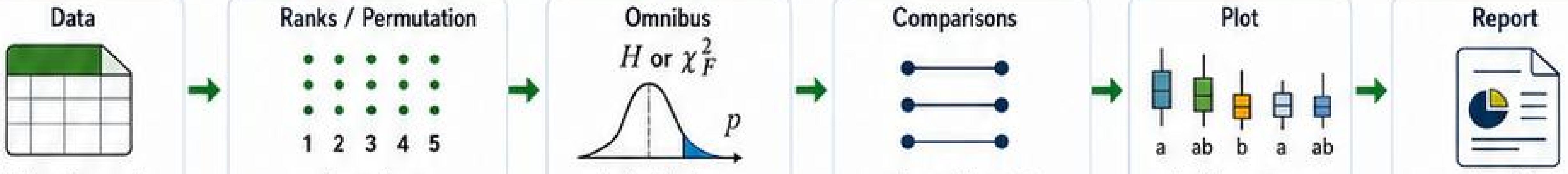
- **np_crd()** — organizes data for CRD
- **np_rcbd()** — organizes data for RCBD
- **agri_rank()** — performs omnibus tests
- **agri_conover()** — Conover post-hoc test (CRD or RCBD/Friedman)
- **agri_pairs()** — pairwise contrasts
- **agri_effects()** — effect sizes, CI and magnitude

4 Interpretation



- Omnibus answers whether there is a global difference among treatments.
- Post-hoc tests show which groups differ from each other.
- Effects and CIs complement the *p*-value and the magnitude of differences.
- In RCBD, blocking reduces experimental noise and increases test power.

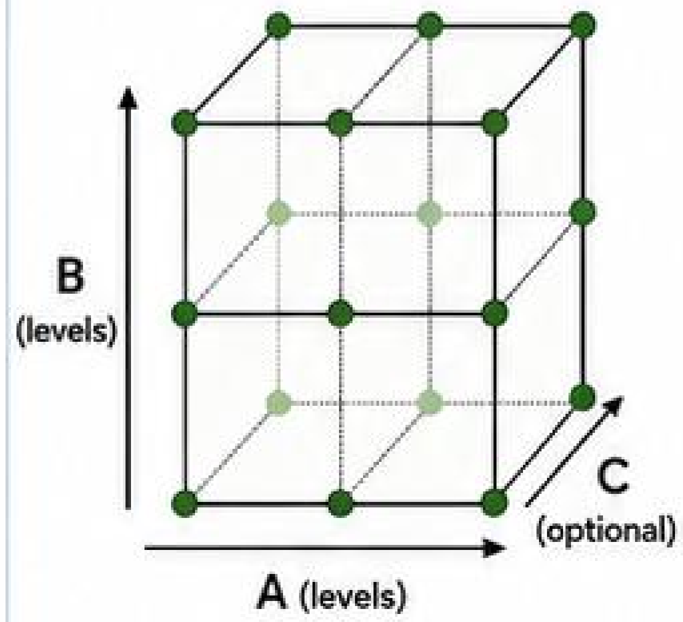
5 Workflow: nonparametric inference (CRD or RCBD)



1. **Data** — collect and organize the data
2. **Ranks / Permutation** — assign ranks or perform permutations
3. **Omnibus** — test global difference
4. **Comparisons** — apply post-hoc tests or pairwise contrasts
5. **Plot** — visualize patterns and groups
6. **Report** — results, effects, CIs and conclusions

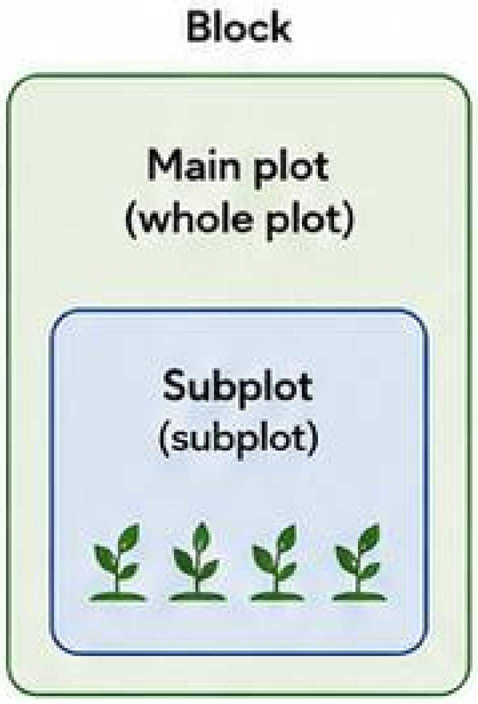
Design an experiment → preserve inference → communicate with clarity. ***That's agriRank.***

1 Factorials



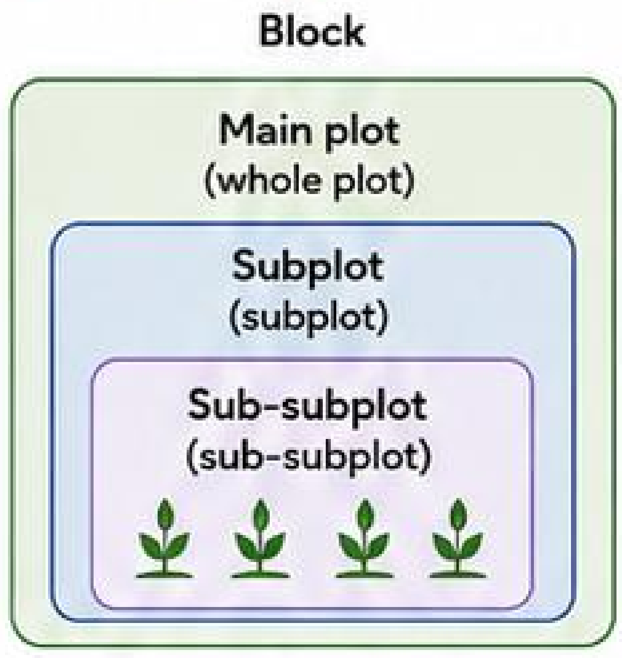
- assess main effects and interaction
- the scientific model must be preserved
- pseudo-ranks and ART are important alternatives

2 Split-plot

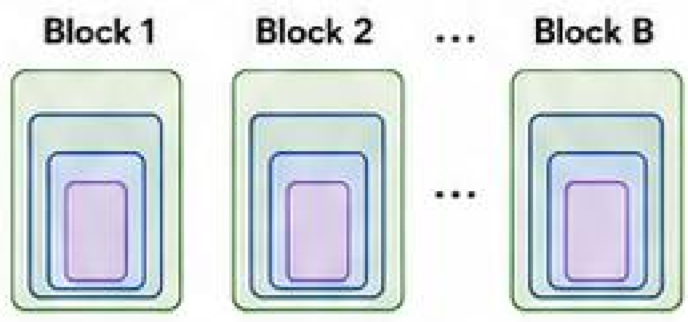


- main plot
- subplot
- two error levels

3 Split-split

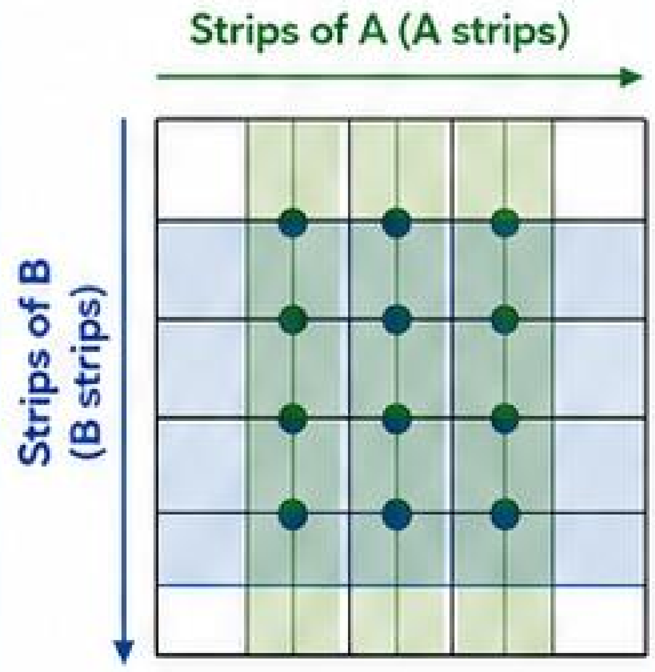


Stratified example



- level 1 (between blocks)
- level 2 (within main plots)
- level 3 (within subplots)

4 Strip-plot



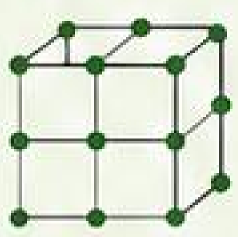
- two series of strips
- A × B intersection
- block × A and block × B structures

5 Analytical engines

Engine	rankFD	ARTool (ART)	permuco
When to use	factorial analyses with independent data	factorial structures with interactions (rank balance)	complex designs with known randomization
Strength	simple, fast and efficient for classic factorial designs	preserves effects and interactions even with non-parametric models and unbalanced data	respects randomization and provides exact p-values or next exact inference
Limitation	does not handle complex plot structures or randomization well	requires correct specification of the model formula and structures	can be computationally intensive for very large samples

6 Functions (np_*)

np_factorial()



complete or fractional factorials

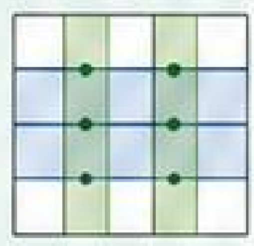
np_splitplot()

plots, main plots and subplots

np_splitsplit()

up to three levels: main plot, subplot and sub-subplot

np_stripplot()



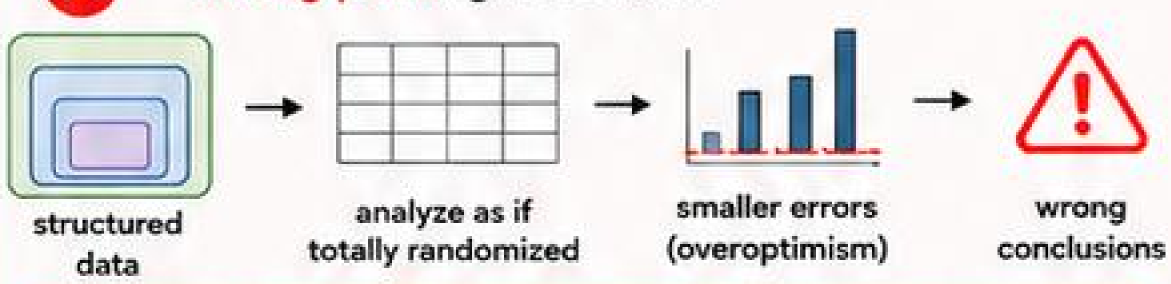
strip-plot (strips) and their intersections

7 Correct reading

- significant interaction changes interpretation
- compare simple effects when necessary
- do not collapse strata
- terms of the formula must be maintained

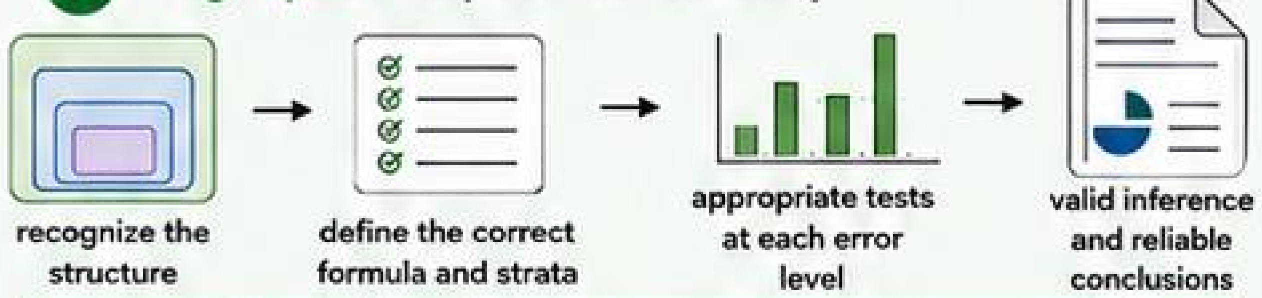
8 Wrong path vs. right path

Wrong path: ignore strata



Ignoring the hierarchy and strata leads to inference errors, inflated error rates and incorrect conclusions.

Right path: respect the hierarchy

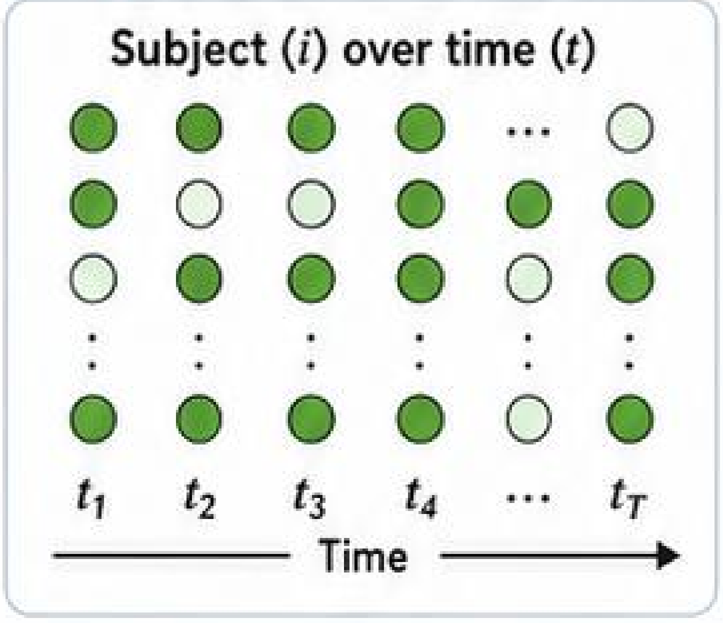


Respecting the hierarchy ensures correct tests, proper control of error type I and coherent interpretations with the experiment.

agriRank | Repeated measures and missing data

Time dependence, subjects, and missingness

1 Repeated measures



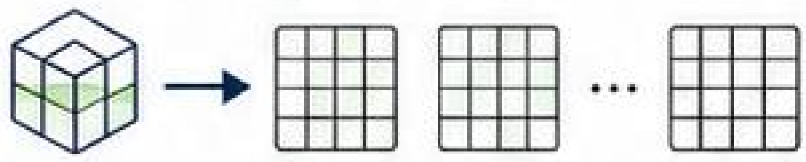
Observed Missing

- same unit observed over time
- within-subject and within-unit are required
- time can interact with treatment
- time dependence must be preserved

Notation

Y_{ijt} $\quad i$ = subject $(1, \ldots, N)$
$\quad j$ = treatment $(1, \ldots, J)$
$\quad t$ = time $(1, \ldots, T)$

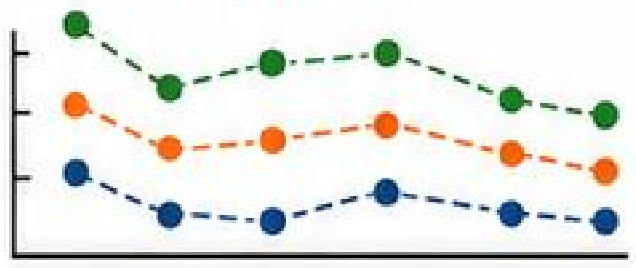
Bootstrap (example)



Resampling residuals *with preservation of the subject-time structure.*

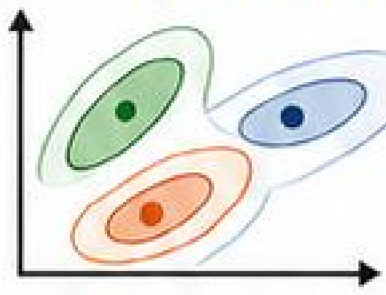
2 Backends

nparLD



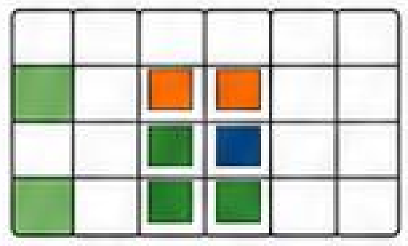
Longitudinal effects nonparametric.

MANOVA.RM



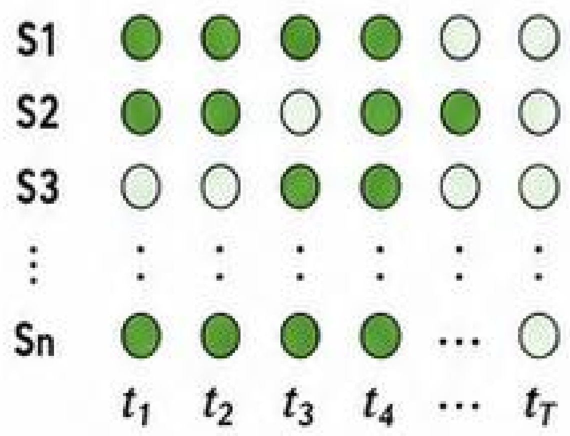
Repeated / multivariate structures based on resampling.

permuco



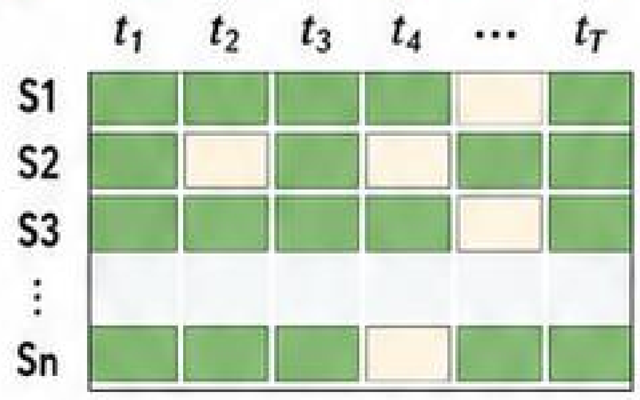
Repeated designs with blocks when admissible.

3 Native engine for nonparametric repeated



- wild bootstrap
- relative effects
- useful for incomplete observations
- *status: experimental*

4 agri_missing_report()

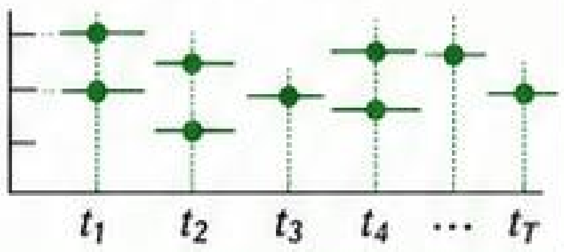


Observed Missing

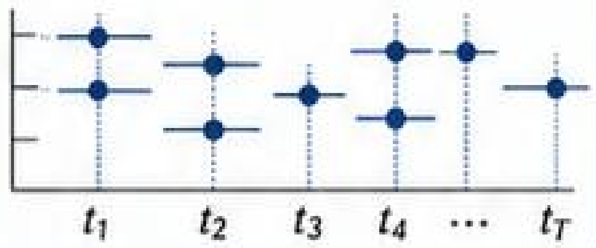
- missing rate *(global and by time)*
- dropout *(accumulated over time)*
- complete / incomplete individuals
- pattern by time *(missing per wave)*

5 agri_missing_sensitivity()

all available



complete cases



Compare results across strategias and assess the stability of conclusions.

6 Safeguards

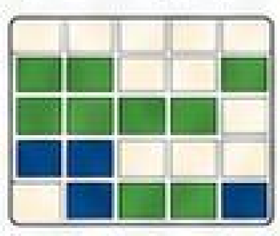
- *do not remove blocks silently*
- *do not treat repeated measurements as independent*
- *do not confuse sensitivity analysis with proof of MCAR.*

7 Recommended workflow

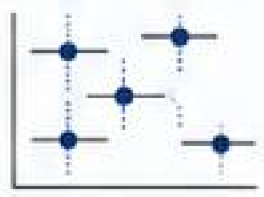
agri_design()

→ agri_repeated() or incomplete_wild_rank_test()

→ agri_missing_report()



→ agri_missing_sensitivity()



→ plots

→ report

Declare the experiment → preserve inference → communicate with clarity. *This is agriRank.*

agriRank | Effects, contrasts and Conover

From global test to interpretable comparison

1 Beyond the p-value

- relative effects
- probability of superiority
- simultaneous CIs when available
- contrasts answer scientific questions

2 Post-hoc tests and comparisons

Dunn

Steel

Conover

Custom contrasts

CLD

A AB B

3 Conover

CRD → **kwAllPairsConoverTest**

```
kwAllPairsConoverTest(
  formula = y ~ trt,
  data = dados,
  p.adjust.method = "holm"
)
```

→ **Full RCBD** → **frdAllPairsConoverTest**

```
frdAllPairsConoverTest(
  formula = y ~ trt + block,
  data = dados,
  p.adjust.method = "holm"
)
```

Incomplete RCBD should not use Conover or Friedman.

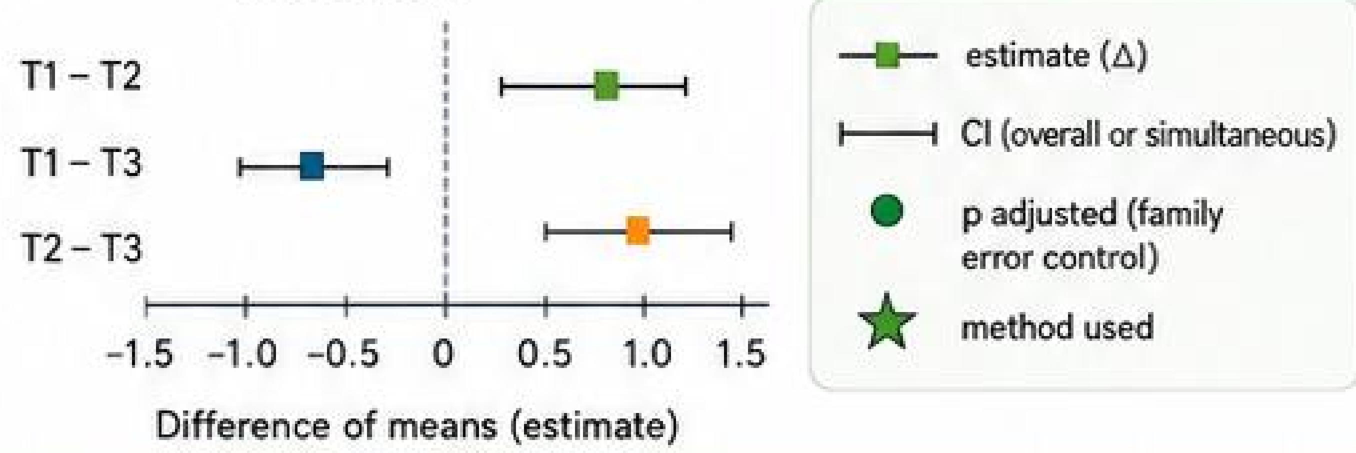
4 Key functions

- **agri_effects()** — estimate relative effects and CI
- **agri_pairs()** — pairwise means and differences
- **agri_conover()** — Conover post-hoc test (kwAllPairsConoverTest or frdAllPairsConoverTest)
- **agri_contrast()** — planned contrasts
- **agri_cld()** — A AB B — group letters (CLD)
- **confint()** — confidence intervals

5 How to read the output

group1	group2	estimate (Δ)	p_value	p_adjusted	method
T1	T2	0.85	0.0014	0.0070	Conover
T1	T3	1.32	0.0002	0.0010	Conover
T2	T3	0.47	0.0210	0.0630	Conover
...	...	...	...	...	...

Forest plot



6 CLD with caution

A AB B

Letters that are the same or share letters do not differ at the specified significance level.

Useful for quick communication, but does not replace estimates and CI.

7 Interpretive workflow

Any overall effect? → Admissible comparisons (post-hoc) → Planned contrasts (specific hypotheses) → Effects and CI (estimable and interpretable) → Final plot (clarity and transparency)

8 Notation for contrasts and relative effects

Pairwise contrast (i vs j)

$$C_{ij}\,\beta = \beta_i - \beta_j$$

Estimate: $\hat{C}_{ij} = \bar{y}_i - \bar{y}_j$

CI: $\hat{C}_{ij} \pm t_{\alpha/2,\,df} \times SE(\hat{C}_{ij})$

Relative effects (probability of superiority)

$$p_i = P(Y_i > Y_j) = P(\varepsilon_i - \varepsilon_j > \mu_j - \mu_i)$$

Nonparametric estimate: $\hat{p}_i = \dfrac{\#(Y_i > Y_j) + 0.5\,\#(Y_i = Y_j)}{n_i \times n_j}$

CI via bootstrap or normal approximation.

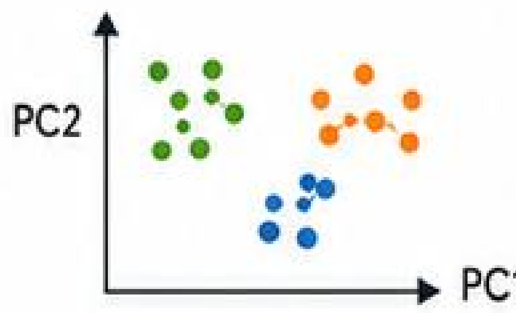
See the agriRank documentation for details on methods and assumptions.

Declare the experiment → preserve inference → communicate with clarity. *That's agriRank.*

agriRank | Multivariate, multi-environment, batch and sensitivity

When there are multiple answers, multiple environments and multiple analytical decisions

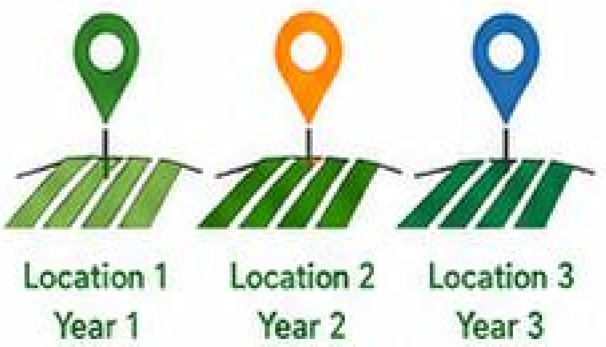
1 Multi-environment



- multiple simultaneous responses
- joint data structure
- integrated classes for tables and reports

Main function: `agri_multivariate()`

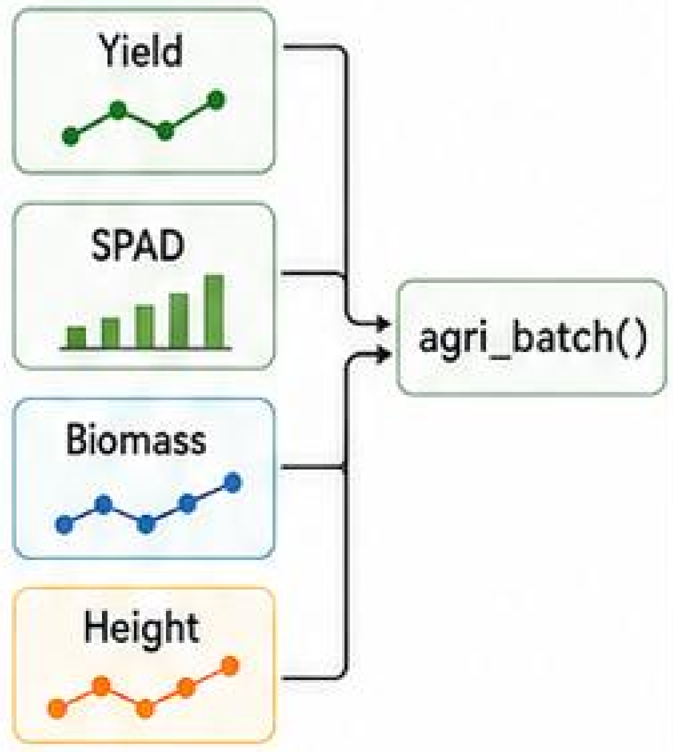
2 Multi-location



- genotypes across locations/years
- environment must not be in the model
- G × E optional
- blocks are specific to each environment

Main function: `agri_multienv()`

3 Batch analysis



- same experimental structure for multiple responses

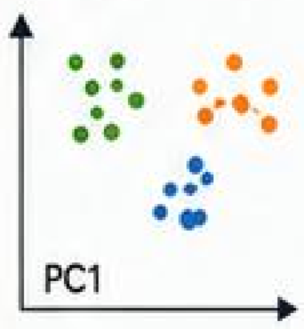
4 Sensitivity analysis

- compares conclusions between methods
- does not focus on the smallest p-value
- assesses the stability of inferences

Main function: `agri_sensitivity()`

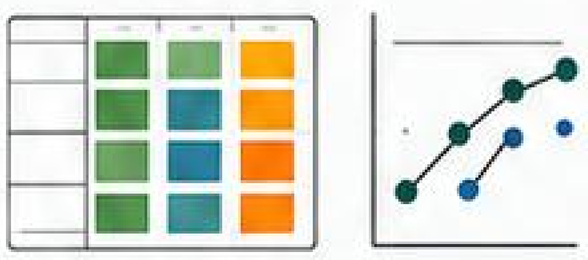
5 Objects and outputs

agri_multivariate_fit

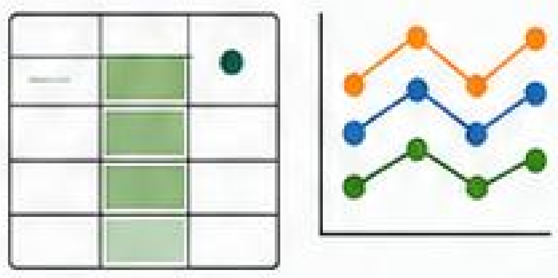


- Scores
- Loadings
- Contributions
- Plots

agri_rank_fit by response



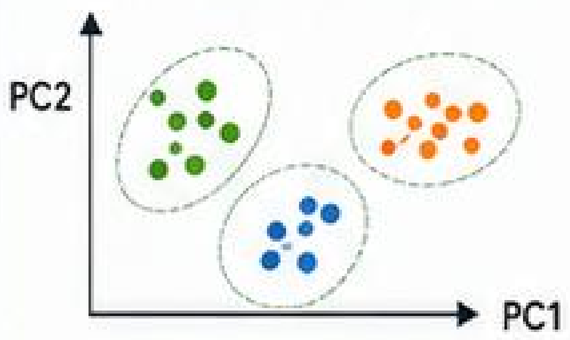
Batch summaries



Exported tables and reports

6 Typical questions

Is there a group effect?

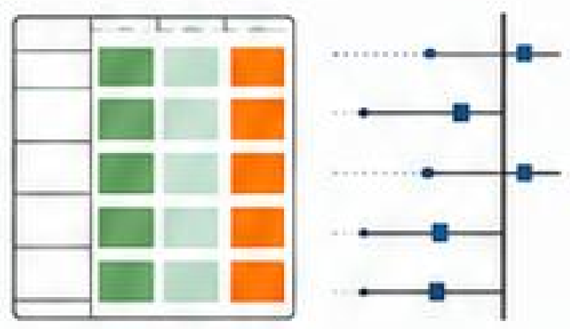


Check whether responses are related and form a common structure.

Is there G × E?

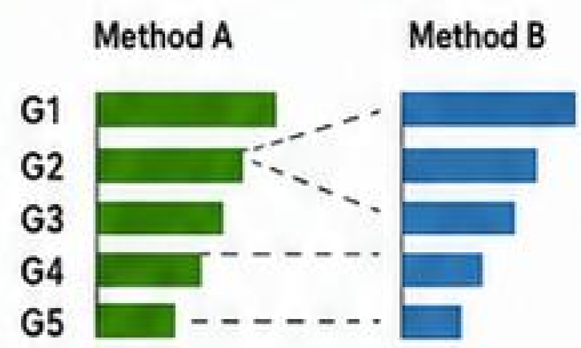
Evaluate whether genotype performance changes across environments.

Are the responses consistent?



Provides agreement or divergence among evaluated responses.

Does the conclusion depend on the method?



Tests the robustness of conclusions when switching the analysis method.

7 Good practices

Declare environment
Always inform the environment and do not model location and year.

Use fixed-effect formulas
y ~ G + E + G:E + block(E)
Include G, E, and if applicable, G × E and block(E) according to the study objective.

Adjust interpretation for multiple responses
Consider loads, correlations, and possible ranking changes.

Save versions and seeds
Record package versions and set.seed() packages for reproducibility.

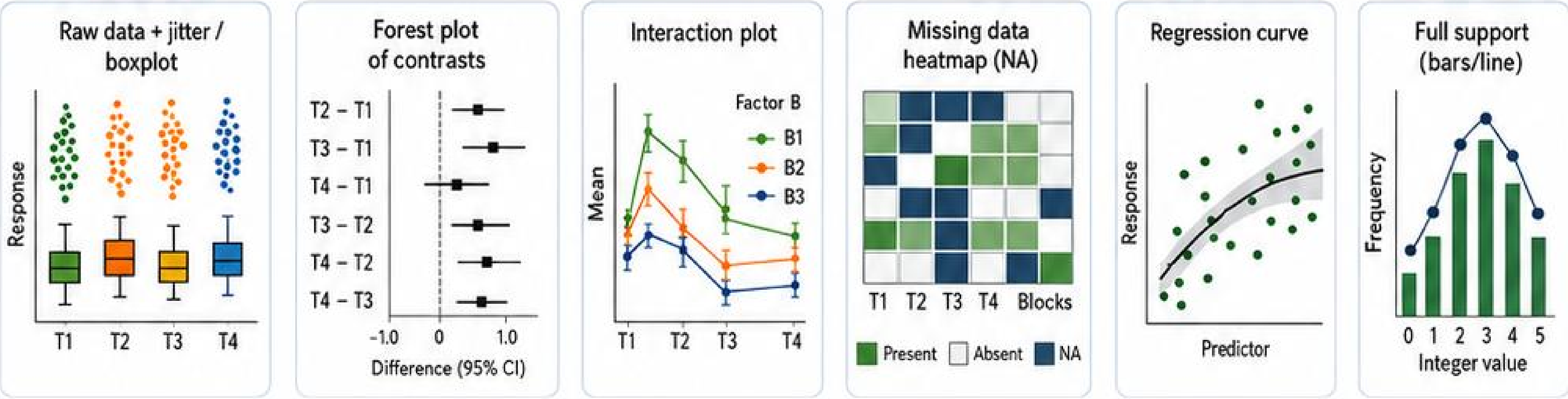
Identify patterns → compare environments → consolidate decisions → communicate clearly. *This is agriRank.*

agriRank | Charts, tables, reports and reproducibility

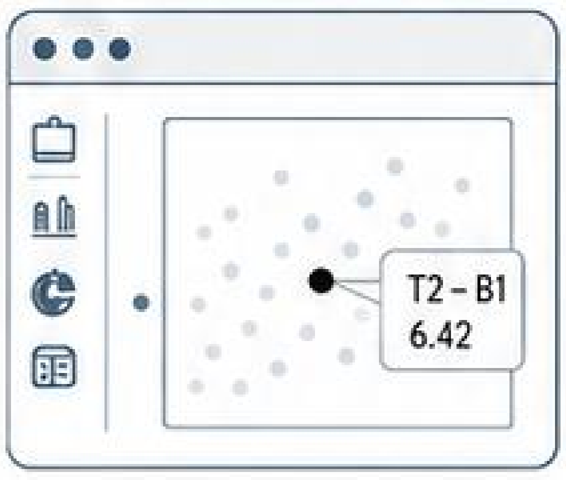
Communicate with clarity and be part of the analysis.

1 Charts

Main functions: agri_plot() and agri_np_plot()



2 Interactivity



Explore with zoom, filters, tooltips and point selection.

Main functions: agri_interactive() and agri_np_interactive()

3 Tables

Main function: agri_table()

Effect table (example ggplot2 style)

effect	estimate	95% CI	p	p-adj
T2 – T1	0.72	[0.28, 1.16]	0.0018	0.0072
T3 – T1	1.35	[0.91, 1.79]	< 0.001	< 0.001
T4 – T1	0.18	[-0.26, 0.62]	0.4210	0.8420
T3 – T2	0.63	[0.19, 1.07]	0.0045	0.0135

- Clear headers
- 95% CI and p-value
- p-adjustment (p.adj)
- Report-ready formatting

4 Reports

Main function: agri_report()

Model object (agri_rank() / agri_np_regression()) → agri_report()

- Study summary
- Results and analysis
- Tables and charts
- Interpretation
- Session information

→ MD Markdown | HTML HTML | DOCX DOCX | PDF PDF

5 Export

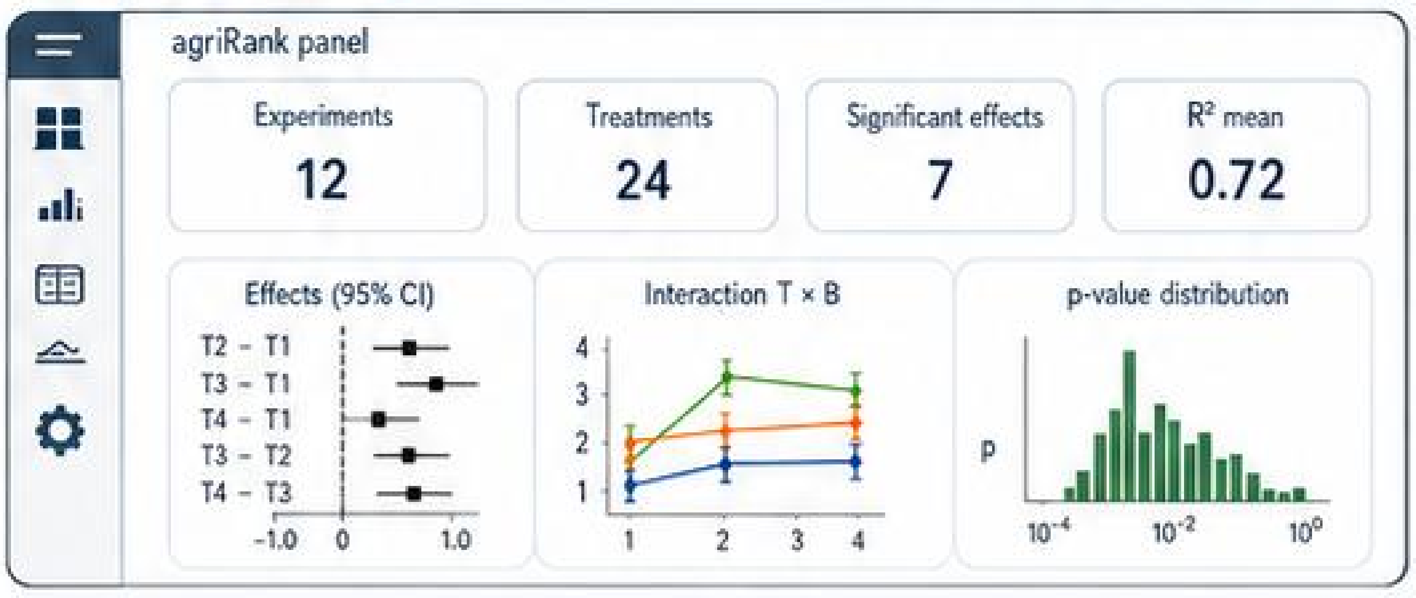
Function: export_results()

- Complete object RDS — RDS
- CSV tables — CSV
- Figures PNG / PDF / SVG — PNG, PDF, SVG

Results ready to share, archive and reuse.

6 Dashboard

Function: agri_dashboard()



7 Reproducibility checklist

1. **seed** — Define and record the seed.
2. **package versions** — Record essential versions.
3. **formula** — $y \sim T * B$ Specify the formula used.
4. **design** — Document blocks, factors and levels.
5. **method/backend** — Record method and backend used.
6. **output files** — Save reports, tables and figures.

8 Key message

The best chart is faithful to the design, shows the observed data and supports inference.

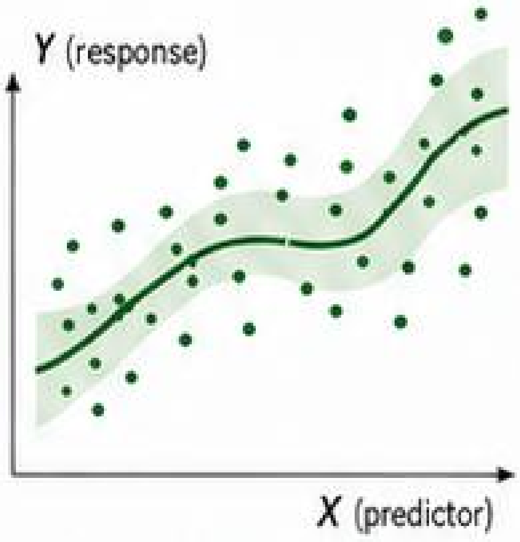
Declare the experiment → preserve inference → communicate with clarity. *That's agriRank.*

agriRank | Nonparametric regression

Continuous relationships without imposing a rigid form

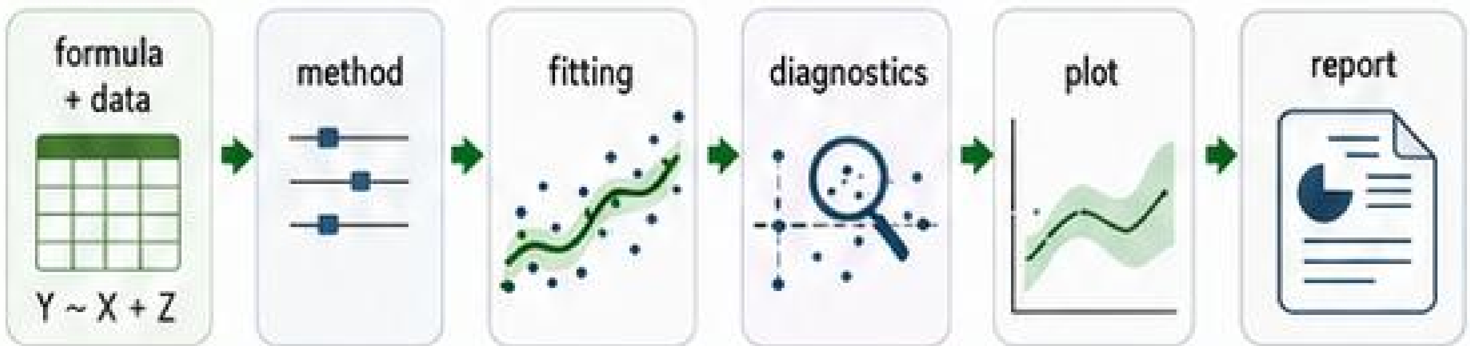
1 When to use?

- quantitative treatments
- environmental gradients
- nonlinear responses
- heterogeneity
- free-form exploration



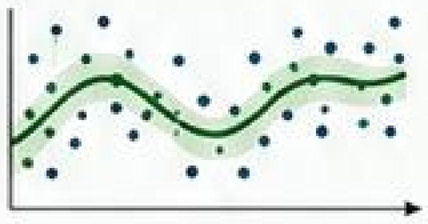
2 Main function

`agri_np_regression()`



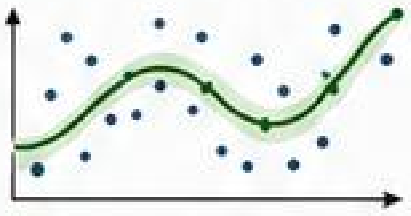
3 Family of methods

LOESS / LOWESS



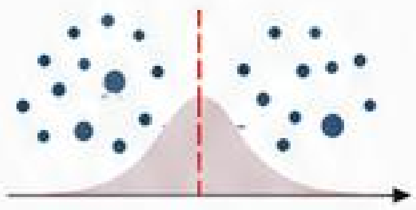
Very flexible; great for local trends and smooth patterns.

Smoothing spline



Global smoothness controlled; good balance for moderate exploration.

Kernel regression

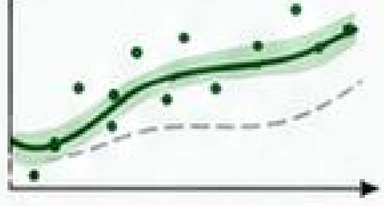


Adaptive and local; captures fine changes based on the data bandwidth.

Isotonic regression

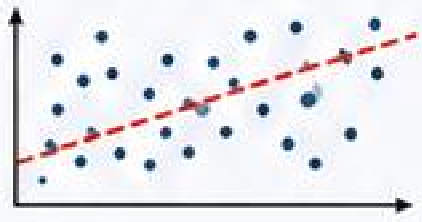
Monotonic by construction; ideal when the relationship must be nondecreasing or nonincreasing.

COBS (Constrained B-spline)



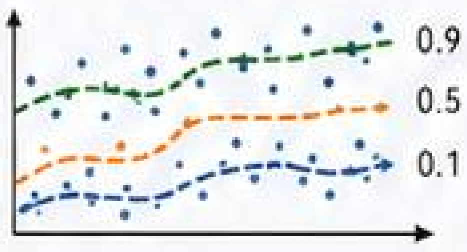
Spline with constraints (monotonicity, convexity); great flexibility with controlled shape.

Theil–Sen regression



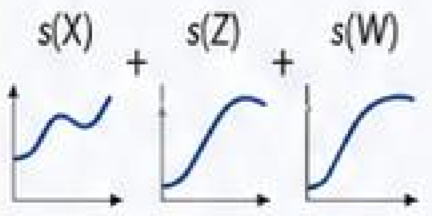
Robust to outliers; estimates slope and median of all pairwise lines.

Quantile regression



Different quantiles (0.9, median, 0.5, 0.1) for robust analysis with outliers.

GAM (Generalized Additive Model)



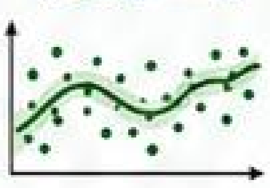
Combines smoothers; models nonlinear relationships with multiple predictors.

SCAM (Shape-Constrained Additive Model)

GAM with shape constraints (monotonicity, convexity, etc.); bio-informed interpretation.

4 Complementary functions

agri_np_predict()



Predictions and uncertainty bands.

agri_np_diagnostics()

Residuals, leverage, QQ plots and graphs.

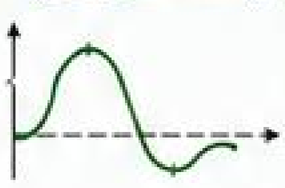
agri_np_compare()

Compare methods by metrics.

agri_np_bootstrap()

Uncertainty via bootstrap and intervals.

agri_np_derivative()



Derivative of curve and change rate.

agri_np_optimum()

Find optimum (max/min).

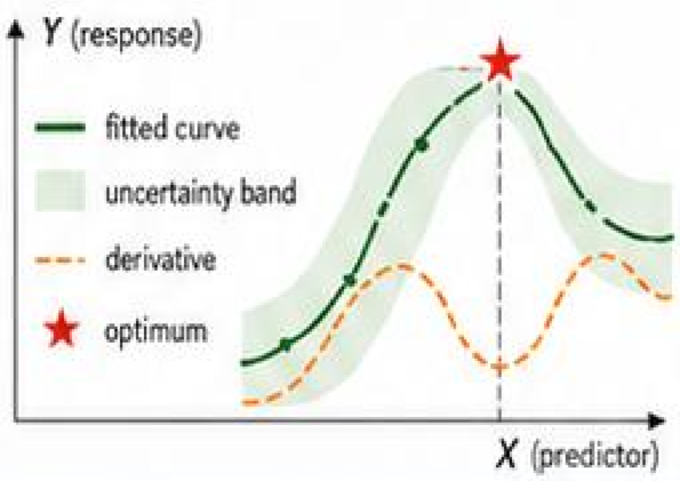
agri_np_significance()

Tests and evidence of effect.

agri_np_specification()

Suggest method and settings.

5 Interpretation



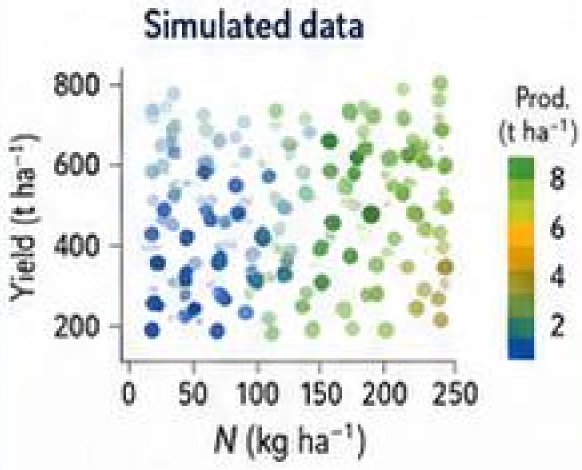
- don't choose method by smallest p
- use predictive error and interpretability
- restrict shape when supported by biological knowledge
- *preserve block when engine permits*

6 Agronomic example

Yield ~ Nitrogen + Rainfall

Compare methods in predicting yield (t ha^{-1}).

Simulated data



Performance (CV5-fold) — lower is better

Method	RMSE (t ha^{-1})	MAE (t ha^{-1})	R^2
LOESS (frac = 0.4)	0.72	0.57	0.74
Smoothing spline	0.69	0.54	0.76
Kernel (gauss, h = opt.)	0.63	0.50	0.80
GAM	0.61	0.48	0.82
SCAM (monotonic N)	0.60	0.47	0.83
Quantile (τ = 0.5)	0.65	0.51	–

7 Key formulas

Nonparametric model

$$y = f(x) + \varepsilon$$

where $f(\cdot)$ is a smooth unknown function and ε is random error.

Regression by quantiles

$$Q_\tau(Y \mid X = x)$$

$\tau \in (0, 1)$ is the conditional quantile of Y given $X = x$ (ex.: $\tau = 0.5$, median).

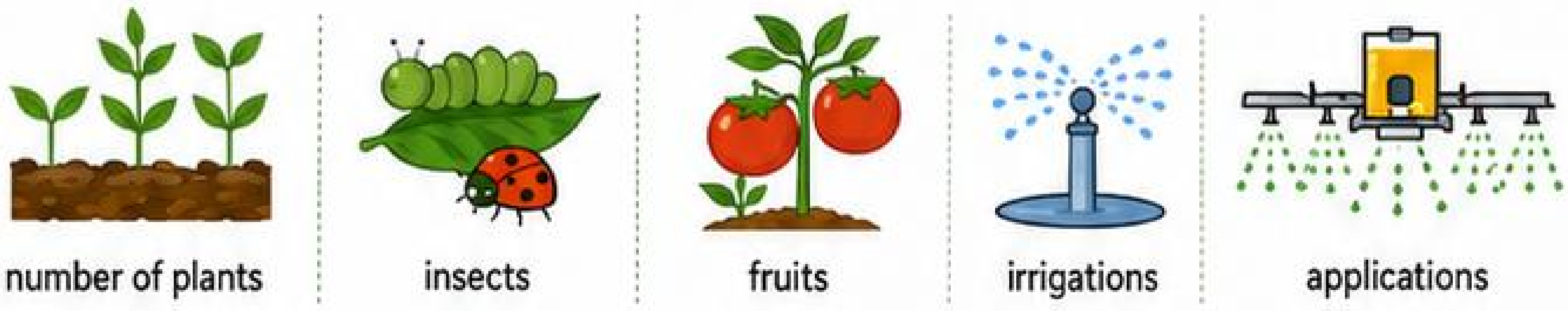
Declare the experiment ⟶ preserve inference ⟶ communicate with clarity. *That's agriRank.*

agriRank | Integer-support regression

When X can only take integer values

1 Examples:

number of plants, insects, fruits, irrigations, applications



2 Key idea

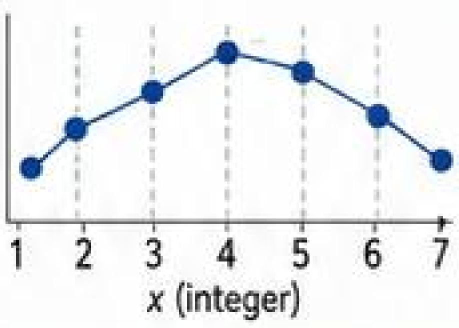
$$x^* = \arg\max_{x \in X_I} m(x)$$

→ decisions must belong to the admissible support

3 Four strategies

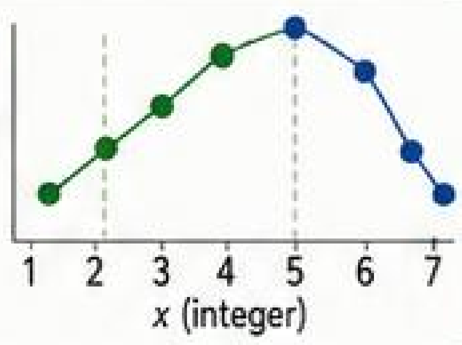
discrete_kernel

kernel to prefer nearby discrete options



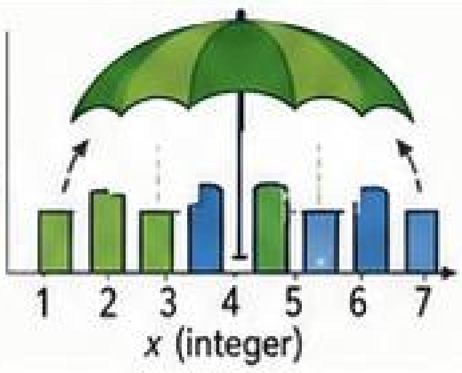
unimodal_isotonic

seeks the maximum and then moves down



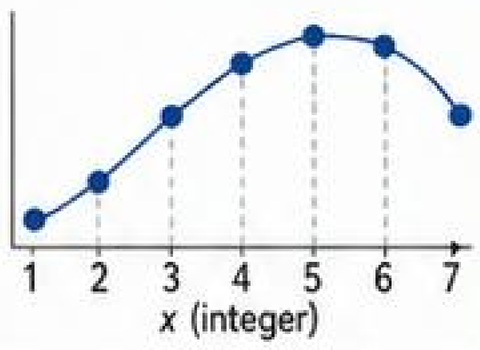
umbrella

exploration in a grid-plus-umbrella with an optional search area



integer_grid

fits a curve between integers and decides on an integer lattice



4 Module functions

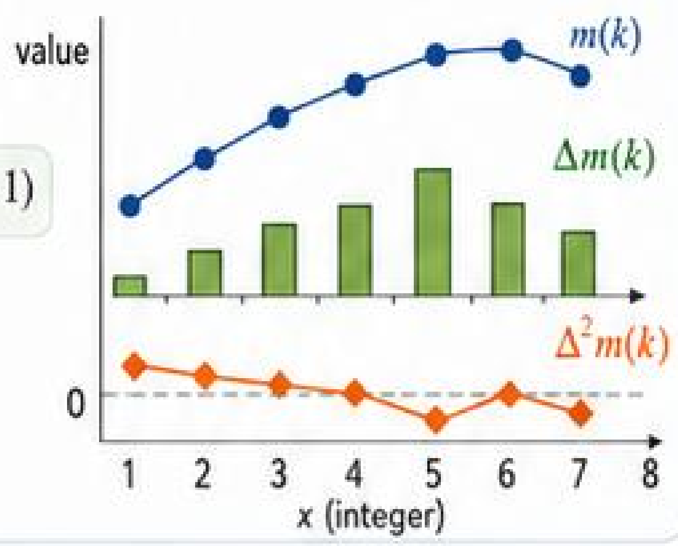
agr_integer_predict()	predict $m(x)$ for each integer x
agr_integer_differences()	first and second finite differences
agr_integer_optimal()	returns the optimal integer x^* and the maximum value
agr_integer_efficiency()	relative efficiency compared to the maximum
agr_integer_threshold()	indicates rates at the inflection point
agr_integer_bootstrap()	bootstrap of discrete curves
agr_integer_confset()	confidence set with 95% coverage

5 Finite differences

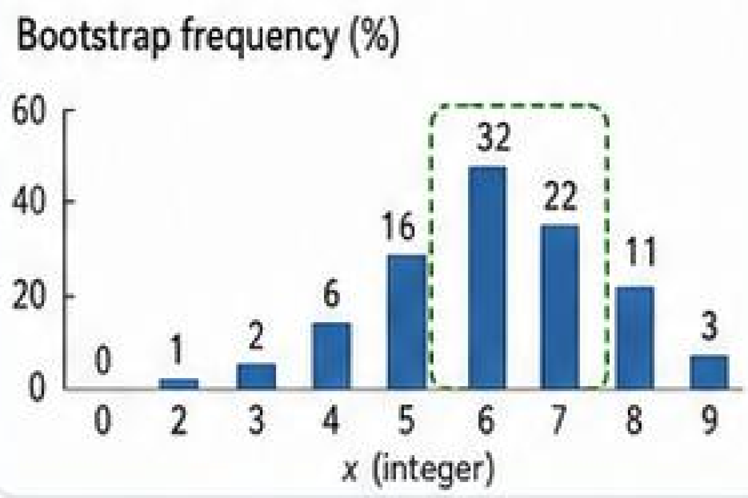
$$\Delta m(k) = m(k+1) - m(k)$$

$$\Delta^2 m(k) = m(k+1) - 2m(k) + m(k-1)$$

- $m(k)$ observed response
- $\Delta m(k)$ first difference
- $\Delta^2 m(k)$ second difference



6 Optimum and uncertainty



- Optimal integer: 5
- 95% confidence: {5, 6, 7}
- 95% of the maximum: 5

7 Safety rules

- avoid extrapolating outside the studied region: the model may be invalid
- ensure support adequacy: the model should be well specified
- avoid information gaps: each block must protect the structure
- avoid repeated measures: the models assume uncorrelated errors

8 In agriRank, operational restriction becomes an advantage.

- wise decisions
- more efficiency
- more yield
- safety and traceability

Declare the *experiment* → preserve the inference → communicate with clarity. ***That's agriRank.***